From demonstration to a defensible answer

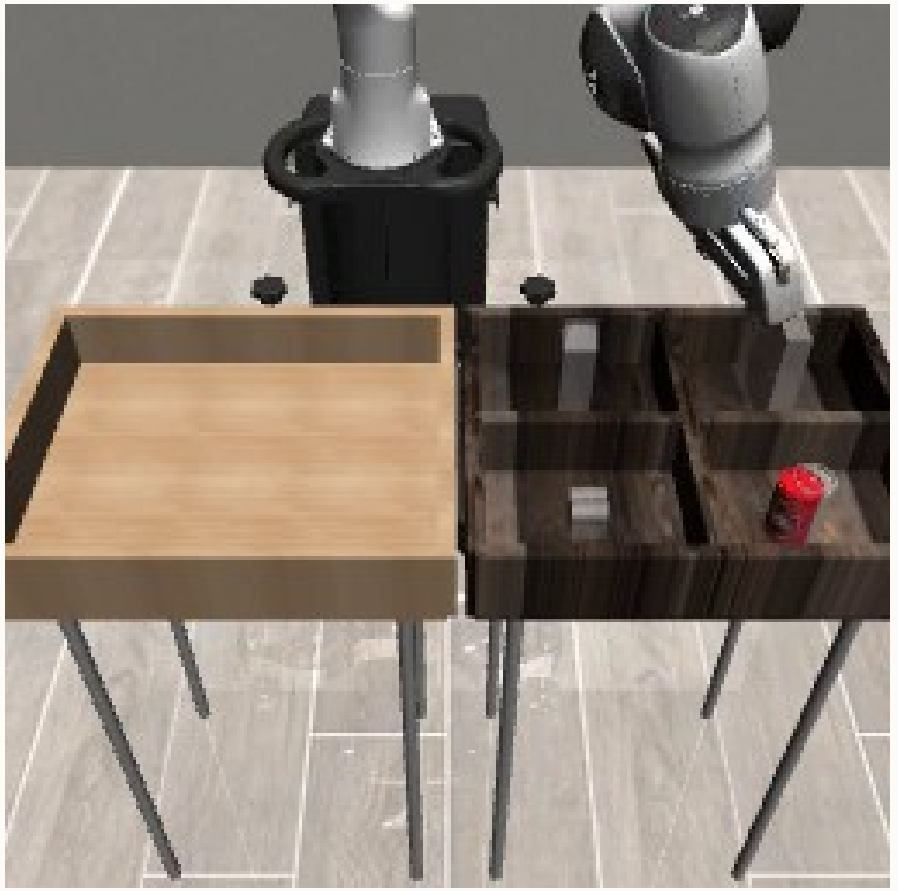
Manipulation Learning Lab

Worked solutions to Q1–Q5

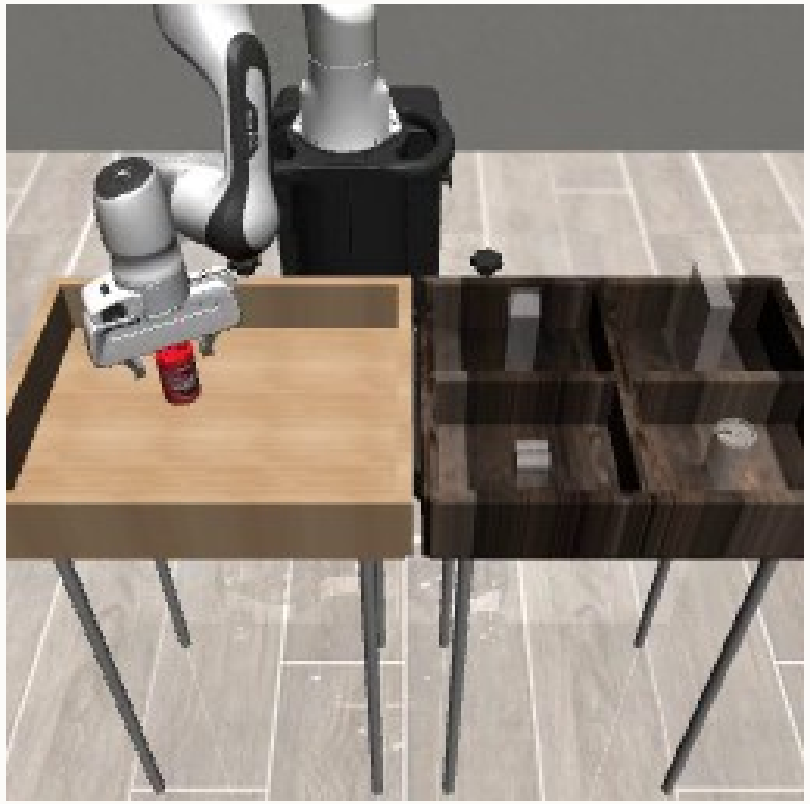
4 executed notebooks

2 policies × 50 epochs

20 real simulator rollouts



What actually counts as task success?



Primary policy lr_a · standard_01

Click to play the recorded rollout

1 Reach and grasp

2 Transport to the target

3 Open the gripper; remove contact

4 Remain stable for 1 second

Stable release: opening ≥ 0.055 m

Displacement ≤ 0.01 m; speed ≤ 0.05 m/s

The policy keeps running during the check.

Q1 · The policy observes 16 state values

Observation	Size	Meaning / units
End-effector position	3	World XYZ, meters
End-effector orientation	4	Quaternion, xyzw
Finger joint positions	2	Two measured slide joints, meters
Can position + orientation	7	World XYZ + xyzw quaternion

3 + 4 + 2 + 7 = 16. The adapter uses object[7:14] in both training and rollout.

Q1 · Seven commands do not mean seven positions

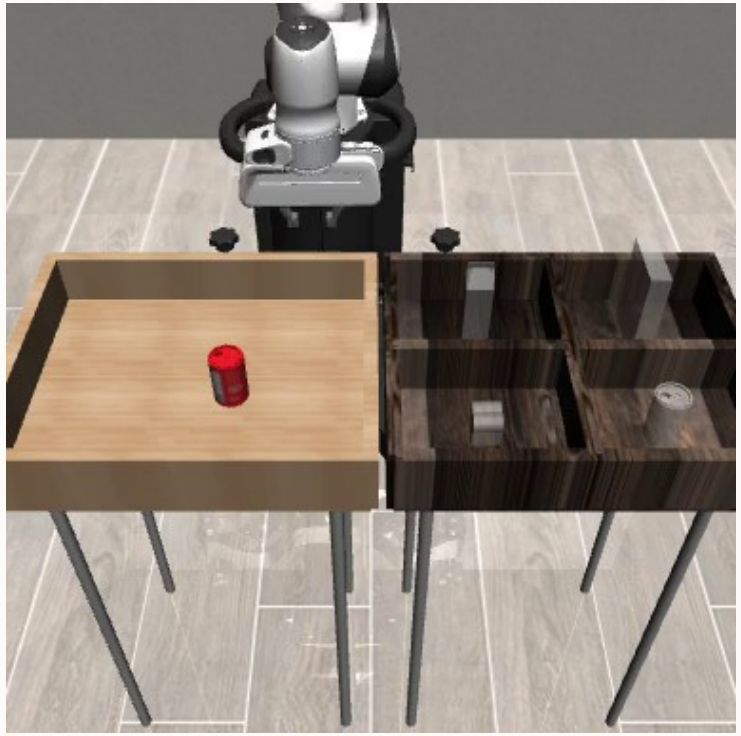
Action channels	Normalized command	Controller interpretation
Translation (3)	[-1, 1]	World increments; limits ±0.05 m
Rotation (3)	[-1, 1]	Axis-angle increments; limits ±0.5 rad
Gripper (1)	Positive / negative	Close / open; unitless

Example: translation command 0.2 requests a 0.01 m controller increment.

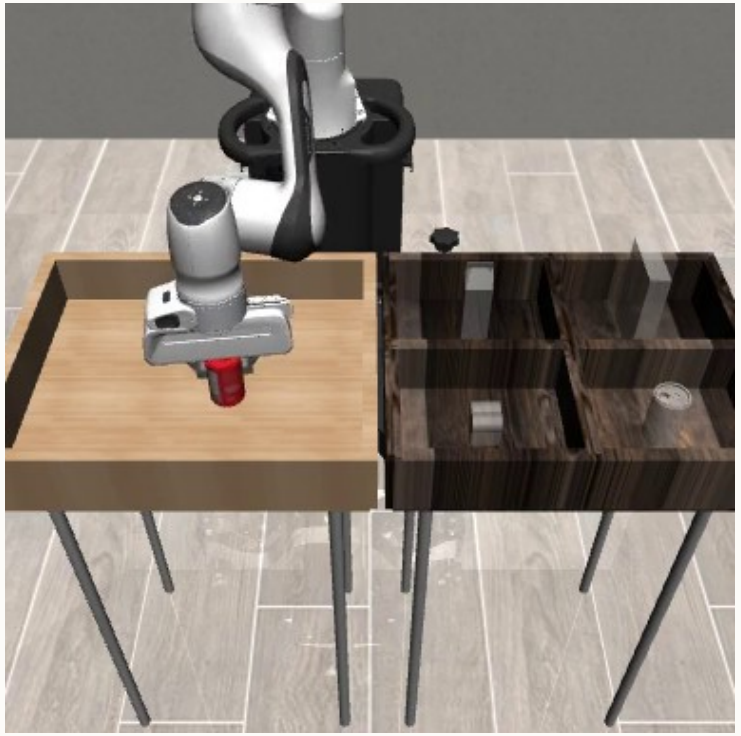
The measured displacement can differ because of dynamics and contact.

Finger joints = 2 / Gripper observation values = 2 / Gripper action = 1 / 20 Hz

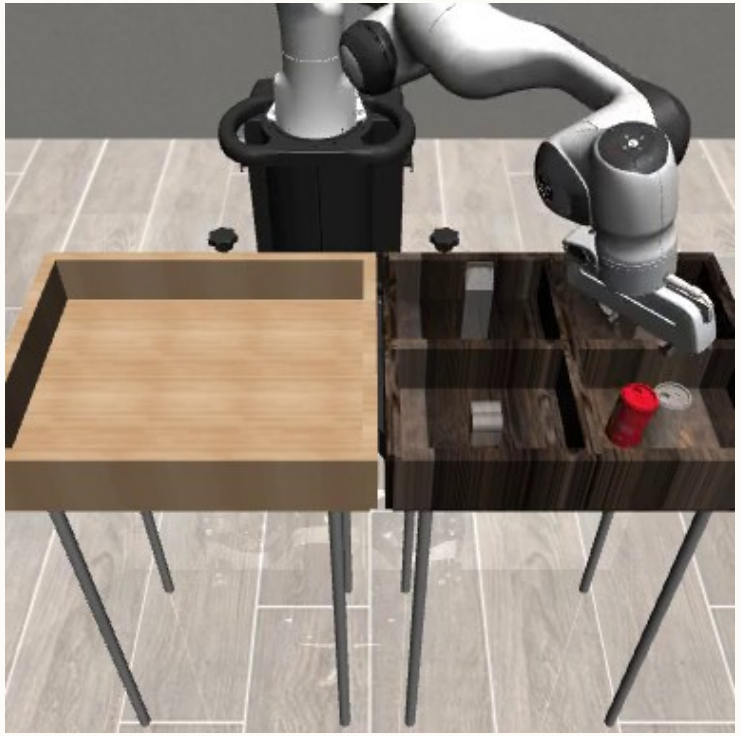
Q1 · A trajectory explains what the final image cannot



Recorded start



Middle of demonstration



Recorded end

Sequence reveals approach, closure, lift and release. Demonstration playback is recorded evidence.

Q2 · Behavior cloning learns the expert action

Observation

16 measured state values

Training-only normalization

→

BC policy

MLP: 256 → 256 → 256

Predict 7 action values

→

Expert target

Recorded action a*

Compare with prediction

$$\text{Loss} = \text{mean of } (\text{predicted action} - \text{expert action})^2$$

200 full episodes: 160 train / 20 validation / 20 test

23,207 recorded control steps; test rollouts never update weights.

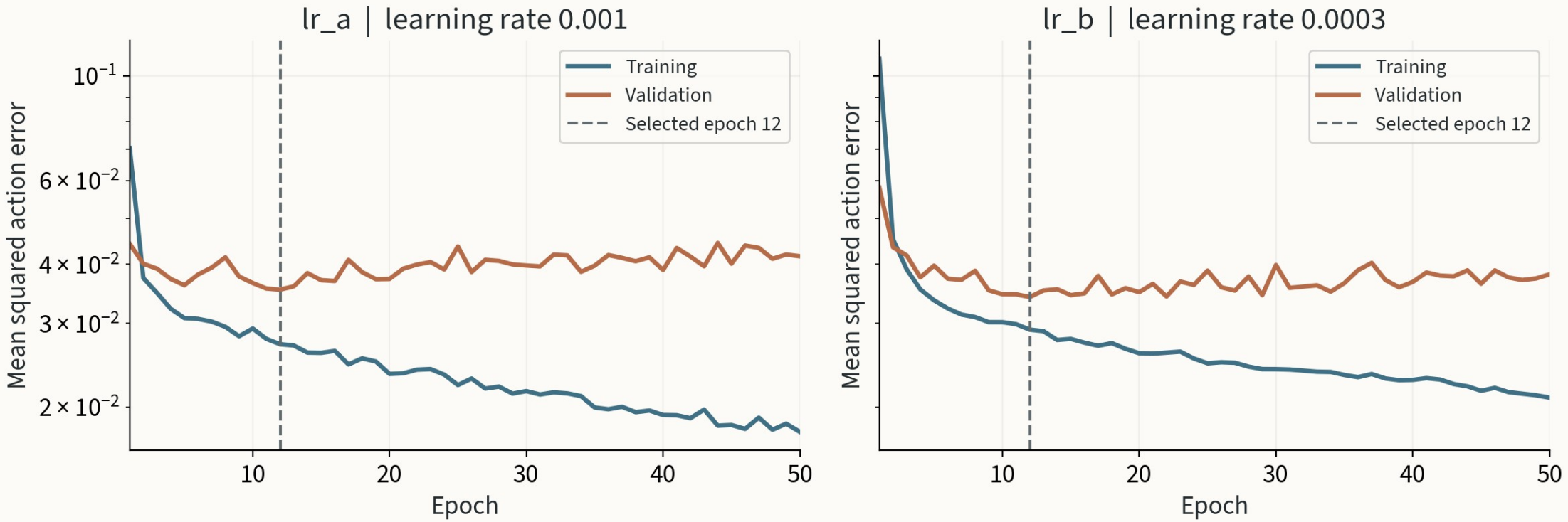
Gradient update learns weights; validation loss selects the checkpoint; test cases measure behavior.

Q2 · Change one variable; keep the comparison fair

Setting	lr_a	lr_b
Learning rate	0.001	0.0003
Seed / epochs / batch	17 / 50 / 256	17 / 50 / 256
Hidden layers	256 / 256 / 256	256 / 256 / 256
Selected validation epoch	12	12
Test cases	Same 5 standard + 5 shifted	Same 5 standard + 5 shifted

All control checks passed. lr_a was fixed as the primary policy before test evaluation.

Q3 · Read training and validation curves together



Best validation MSE: lr_a = 0.03538; lr_b = 0.03412. Both trained for 50 epochs.

Q3 · Lower training error did not yield more successes

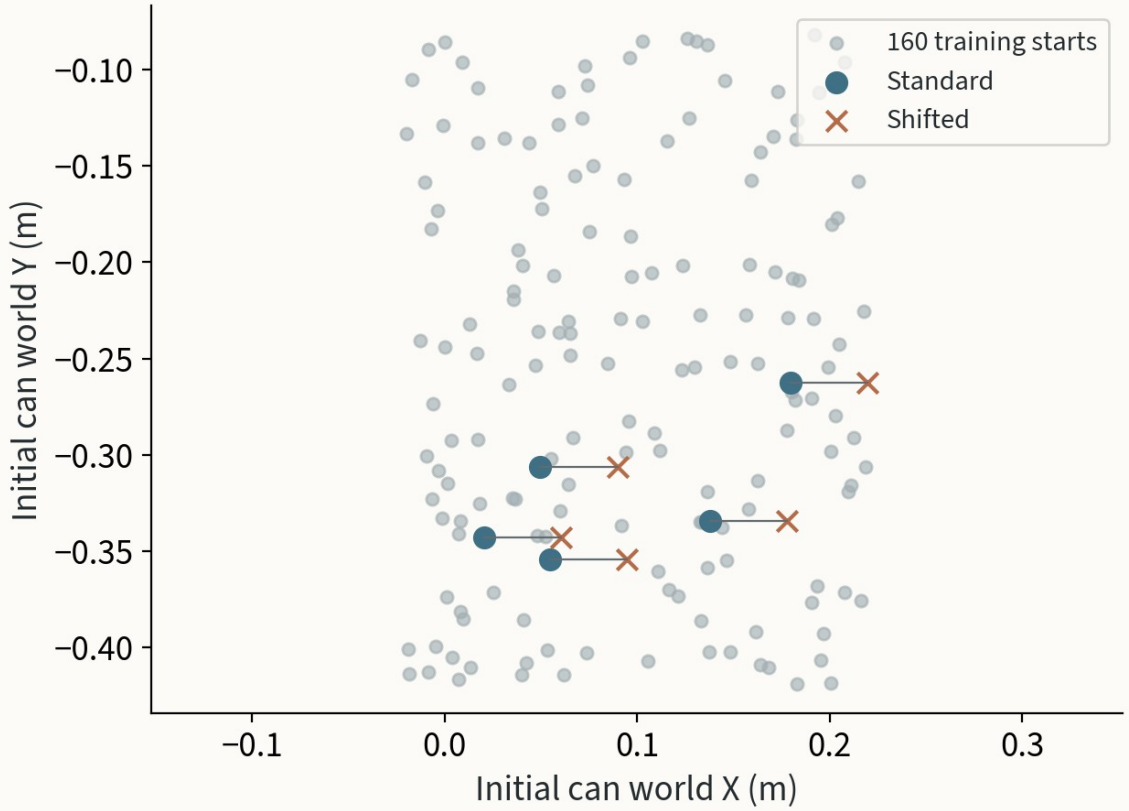
Variant	Epoch-50 train	Selected val.	Standard	Shifted
lr_a	0.01772	0.03538	3/5	3/5
lr_b	0.02092	0.03412	5/5	5/5

A rollout changes the states the policy will see next.

Small action errors can accumulate; a missed grasp changes the rest of the task.

Counts use stable released placement. Environment-success counts happen to match in this run.

Q4 · Position Shift changes only the initial can position



Standard start → +0.04 m in world X

Same source state and evaluation seed

Shifted can stays inside the source bin

Dots are initial positions.

Lines connect paired starts.

Neither marks a trajectory or success.

2D overlap and a four-centimeter shift do not certify out-of-distribution generalization.

Q4 · Same success labels, different completion times

Source / seed	lr_a std.	lr_a shift	lr_b std.	lr_b shift
demo_39 / 101	Fail · 500	Fail · 500	Pass · 144	Pass · 145
demo_185 / 102	Pass · 141	Pass · 156	Pass · 134	Pass · 142
demo_29 / 103	Pass · 157	Pass · 163	Pass · 149	Pass · 153
demo_48 / 104	Fail · 500	Fail · 500	Pass · 182	Pass · 141
demo_157 / 105	Pass · 150	Pass · 159	Pass · 143	Pass · 142

Cells show result · control steps. lr_a / demo_185: 141 → 156 steps (+0.75 s); both succeed.

Q5 · The arm moves on without the can



lr_a · standard_00 · 500 steps / 25 s

Click to play the unmodified failure video

Contact is present at steps 49–56.

At step 57, contact becomes false.

The gripper still receives a close command.

The arm rises; the can stays behind.

Failure stage: grasp retention / start of lift

Annotation: first supported loss of contact

Root cause: still a hypothesis

Q5 · Pin the explanation to recorded steps



Step 45 / frame 11

Opening 0.0795 m

Can Z 0.8601 m

Frame offset +0 steps

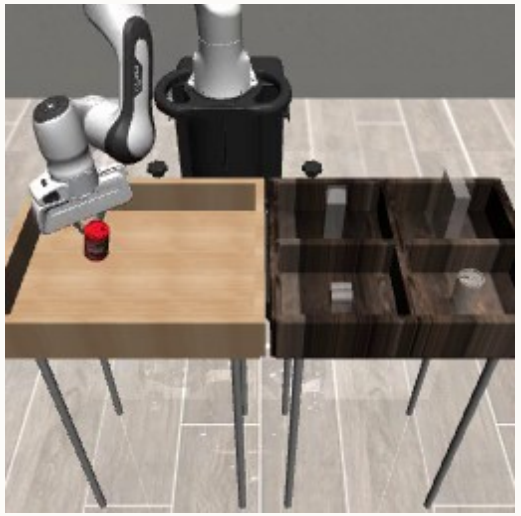


Step 53 / frame 13

Opening 0.0534 m

Can Z 0.8599 m

Frame offset +0 steps

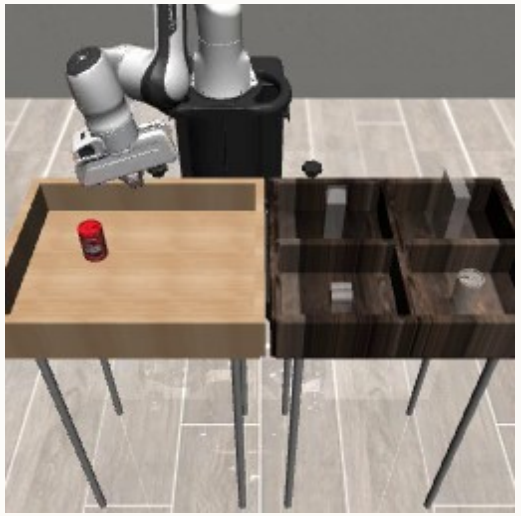


Step 57 / frame 14

Opening 0.0224 m

Can Z 0.8647 m

Frame offset +0 steps



Step 65 / frame 16

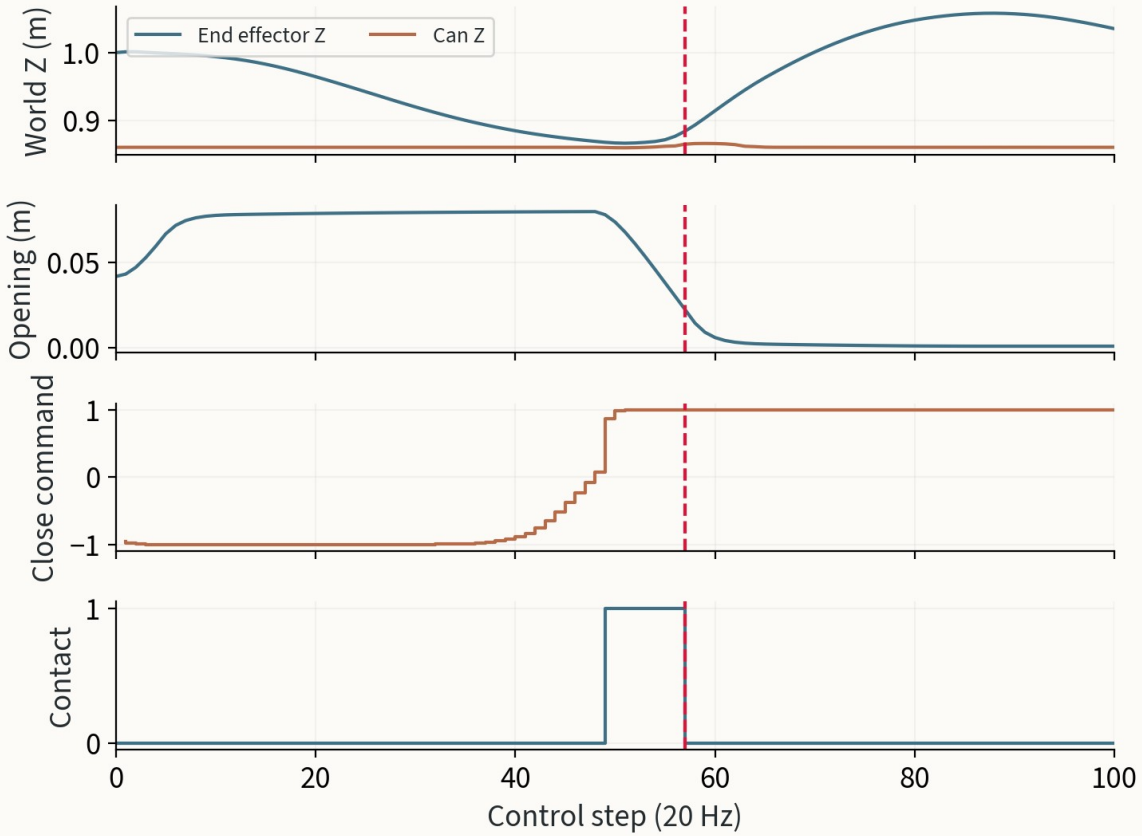
Opening 0.0022 m

Can Z 0.8603 m

Frame offset +0 steps

a[t−1] leads to s[t]. Frames are sampled after actions: steps 1, 5, 9, …; use the saved mapping.

Q5 · Separate measured facts from a causal story



Observed

Contact lost at step 57

Close command ≈ +1

Opening: 0.02240 m

Maximum can lift: 5.72 mm

Hypothesis and test

Grasp alignment or closure timing.

Test one timing intervention from the same saved pre-close state.

The red line is an evidence-based annotation. The proposed intervention has not been executed.

A complete solution is a chain of evidence

Question	Evidence to include	What the answer establishes
Q1	Data contract + demonstration	Observation, action, units and sequence
Q2	Configuration + selected checkpoint	How the mapping is learned
Q3	Both loss histories + success counts	Prediction loss versus task behavior
Q4	Paired rows + initial-position plot	Observed sensitivity; limited coverage
Q5	Trace + video + annotated steps	Failure fact, hypothesis and test

One training seed; five cases per condition; simulation only. Retain failures and the limits of inference.